

DELIVERY-AWARE SPARSE CAUSAL ROUTING FOR DISTRIBUTED MULTI-OBJECT TRACKING

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ABSTRACT

We study sparse communication for distributed labeled multi-Bernoulli tracking when mobile formations change their physical neighbors. A causal routing policy retains a feasible formation tree, repairs it only when necessary, and combines local directed cycles with bidirectional gateways. The scheduled backbone uses $N+2(F-1)$ posterior messages for N sensors in F formations. We distinguish this planned graph from packet delivery and label-specific fusion, and derive a finite-round sensitivity bound for exact geometric fusion. Paired 160-step development experiments with 24 and 36 sensors reduce attempted posterior bytes by 10.0% and 21.8% relative to fixed routing. Conditional localization RMSE decreases substantially, but set-error gains are smaller and a consistency tradeoff remains at 36 sensors. A post-hoc error budget shows that fixed-cardinality localization improvements alone can reduce mean OSPA by at most 0.28 and 1.31 m in the sparse arms. A receiver ablation exposes a precision-consistency tradeoff, reinforcing target-set recovery as the unresolved objective.

Index Terms— Multi-object tracking, labeled multi-Bernoulli, distributed fusion, dynamic topology, packet loss

1. INTRODUCTION

Mobile sensor formations exchange local posteriors to track objects beyond individual fields of view. As formations move, an initial route can lose an inter-formation connection even when an alternative physical link exists. Retaining many inputs improves redundancy but repeatedly transmits multi-object densities. The design problem is which posteriors reach which receivers within a finite tracking interval.

The labeled multi-Bernoulli (LMB) filter represents object existence and labeled spatial states [1]. Consensus LMB filtering uses Kullback–Leibler averaging (KLA), or normalized geometric fusion [2, 3]. Classical consensus analyses repeatedly fuse fixed inputs through an appropriate stochastic matrix. Mobile tracking instead inter-

leaves new observations with few communication rounds. Scheduled connectivity then guarantees neither delivery nor lower tracking error.

Randomized gossip relates averaging time to a stochastic matrix spectrum [4]; event-triggered Bayesian filtering uses posterior divergence from a predicted last transmission [5]. We instead study route repair under changing physical feasibility and the inputs actually received for finite-round multi-object fusion.

Multi-view GCI addresses label absence under unequal fields of view [6]. Here that fusion approximation is held fixed while communication structure and missing-input handling are compared. Routing uses physical links, not posterior-dependent learned weights.

This paper develops a sparse causal backbone and analyzes how packet loss changes its intended fusion. The contributions of the present study are:

- a formation-level repair rule with an exact message count under a local-cycle and bidirectional-tree architecture;
- a separation of scheduled, packet-delivered and label-active graphs, with a finite-round perturbation bound for exact geometric fusion;
- paired fixed, full-repair and sparse-repair experiments exposing both the communication gains and the remaining set-estimation tradeoffs.

All results use one opened seed per scale and constitute development evidence, not statistically established cross-scenario superiority.

2. SPARSE CAUSAL LMB ROUTING

2.1. Fusion and communication model

Sensor i maintains an LMB posterior $\pi_i = \{(r_i^\ell, p_i^\ell)\}_{\ell \in \mathbb{L}_i}$, where r_i^ℓ is the existence probability and $p_i^\ell(x)$ is a normalized spatial density. For a common active label and

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weights $w_j \geq 0$, $\sum_j w_j = 1$, exact KLA satisfies [3]

$$\eta^\ell = \int \prod_j p_j^\ell(x)^{w_j} dx, \quad \bar{p}^\ell(x) = \frac{\prod_j p_j^\ell(x)^{w_j}}{\eta^\ell}, \quad (1)$$

$$\text{logit } \bar{r}^\ell = \sum_j w_j \text{logit } r_j^\ell + \log \eta^\ell. \quad (2)$$

Thus changing a neighbor affects both spatial overlap and existence. More mixing need not preserve the estimated target set.

Let G_t^{phys} describe current physical links, and let A_t be the scheduled directed graph, with rows denoting receivers and columns senders. The controller observes current positions, link reliabilities and its previous route, but no target truth or future trajectory. The implemented controller uses network-wide current geometry; decentralized acquisition and dissemination of this control information are outside the measured posterior-payload budget. Fusion itself is performed at each sensor.

2.2. Causal tree repair and sparse inputs

Sensors are partitioned into F formations. A candidate edge in the formation graph is usable when current physical sensor pairs can realize the required gateways. The policy first tries the preceding formation tree. When that tree or its gateway assignment is infeasible, it selects a current feasible tree lexicographically: minimize replaced formation edges, maximize bottleneck reliability, maximize total log reliability, then minimize distance. Gateway feasibility includes the receiver input constraints. The retained object is the formation tree, not every sensor-level edge: local cycles and gateway endpoints are rebuilt from current geometry.

The sparse policy retains one directed cycle inside each formation. Each undirected edge of the formation tree is realized by one directed gateway in each direction; the two directions need not use the same sensor pair. Every sensor receives one dominant local input, and gateway receivers also receive one cross-formation input. The planned weights are 0.70 for the dominant input and 0.05 for a retained residual input, with the remaining mass assigned to self. The full-repair reference instead retains two nonself inputs per receiver. Removing its unused local residual input increases the sparse receiver's self weight by 0.05.

Proposition 1 (scheduled backbone). If every local cycle and every directed gateway is physically feasible, the scheduled sensor graph is strongly connected and contains exactly

$$M = N + 2(F - 1) \quad (3)$$

directed messages per round. Proof. The local cycles contribute N edges and provide directed paths between

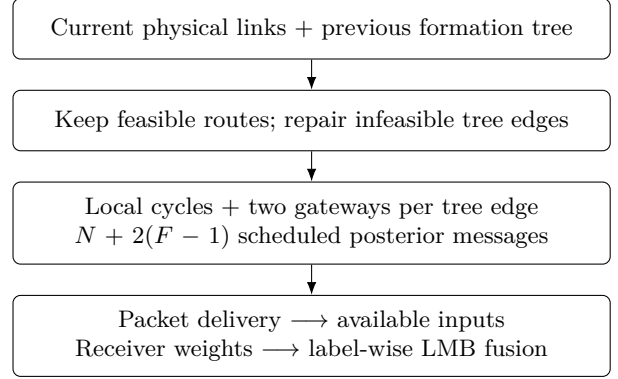


Fig. 1. The routing and fusion pipeline. Scheduled connectivity is a property of the third stage; actual information use also depends on delivery and label availability in the fourth stage.

sensors in each formation. The tree contributes $F - 1$ formation edges, each realized in both directions. Concatenating local paths and gateways gives a directed path between any two sensors. The edge sets are disjoint, giving (3). This is a count for the chosen architecture, not a global lower bound over all strongly connected directed graphs or a guarantee about delivered links.

3. PACKET DELIVERY AND FINITE-ROUND FUSION

3.1. Where a missing weight goes

Let W be a planned row-stochastic matrix with positive diagonal. For receiver i , let $D_{ij} \in \{0, 1\}$ indicate delivery, set $D_{ii} = 1$, and define missing weight mass $m_i = \sum_{j: D_{ij}=0} W_{ij}$. The current reference implementation omits an unavailable input and uses

$$R_{ij} = \frac{W_{ij} D_{ij}}{1 - m_i}. \quad (4)$$

This differs from the planned removal of a local edge. An existing alternative receiver rule returns missing mass to self:

$$S_{ij} = W_{ij} D_{ij} + m_i \mathbf{1}\{i = j\}. \quad (5)$$

Both are row stochastic, and both satisfy $\|R_{i:} - W_{i:}\|_1 = \|S_{i:} - W_{i:}\|_1 = 2m_i$. Their distinction is therefore not a smaller row-wise perturbation norm. Rule (5) preserves the planned weight of each surviving neighbor. If a 0.70-weight dominant input is missing, a surviving 0.05-weight input becomes $0.05/0.30 = 0.1667$ under (4), but stays at 0.05 under (5). The latter avoids amplification but can slow useful information transfer.

Neither rule preserves double stochasticity under arbitrary directed losses. For example, a symmetric two-node matrix with off-diagonal weight a becomes $\begin{bmatrix} 1 & 0 \\ a & 1-a \end{bmatrix}$

after one direction fails under self fallback. Its column sums are $1 + a$ and $1 - a$. Reciprocal scheduled gateways alone consequently do not establish equal-weight KLA consensus after packet loss.

3.2. A finite-round sensitivity bound

Consider exact geometric fusion of fixed positive densities on a common support, without intervening observations:

$$q_\alpha(z) = \frac{\prod_j q_j(z)^{\alpha_j}}{\int \prod_j q_j(u)^{\alpha_j} du}, \quad \alpha \geq 0, \quad \mathbf{1}^\top \alpha = 1. \quad (6)$$

Assume $\text{osc}(\log q_j) = \sup_z \log q_j(z) - \inf_z \log q_j(z) \leq L$. For probability vectors α, β ,

$$D_{\text{KL}}(q_\alpha \| q_\beta) \leq L \|\alpha - \beta\|_1. \quad (7)$$

Let $P = \widetilde{W}_K \cdots \widetilde{W}_1$ and $Q = W_K \cdots W_1$ be delivered and planned products. Then

$$\max_i \|P_i - Q_i\|_1 \leq \sum_{k=1}^K \|\widetilde{W}_k - W_k\|_\infty \leq 2 \sum_{k=1}^K \max_i m_{i,k}. \quad (8)$$

For (7), $h = \sum_j (\alpha_j - \beta_j) \log q_j$ has oscillation at most $L \|\alpha - \beta\|_1$; the log normalization ratio lies between $\inf h$ and $\sup h$. Equation (8) follows by telescoping and $\|W_k\|_\infty = \|\widetilde{W}_k\|_\infty = 1$. These elementary bounds favor neither fallback rule and guarantee neither lower tracking error nor uniform averaging. Unbounded Gaussian support, mixture truncation, changing labels and Bayesian updates prevent applying them directly to the complete recursive tracker.

4. EXPERIMENTS AND REMAINING TRADEOFFS

4.1. Paired protocol

We use a temporally coupled formation-braid simulation with four or six formations of six sensors, denoted M24 and X36. The scenes contain 16 and 24 targets over 160 time steps. Sensors within a formation share a heading, with a 120° field of view and 300 m sensing range. Detection probability is nominally 0.88, clutter rate is 4 and the base measurement noise standard deviation is 7 m. Detection probability decreases and measurement covariance increases with range and off-axis angle; these state-dependent sensing settings are shared by every arm. Target handovers overlap periods of physical route failure. Each scale uses seed 1301; truth, measurements, directed delivery uniforms and filter randomness are paired across its arms.

The fixed baseline keeps its initial formation tree, while recomputing sensor gateways when that tree remains feasible. Otherwise, it retains the still-physical

edges of its initial sensor route. Full causal repair retains two inputs per receiver and changes the formation tree when required. Sparse causal repair uses (3). These three arms share packet renormalization (4). A fourth X36 arm changes only missing-input handling to (5), including unavailable or empty neighbor packets.

All arms use a componentwise powered-Gaussian-mixture approximation to KLA: at most three input components, eight fused components and 256 component tuples. This preserves multiple modes but is not an exact arbitrary-GM power. Label-absence handling is held fixed across arms. The implementation uses one communication/fusion round per tracking step.

We report position-only Euclidean OSPA (E-OSPA), with order 2 and cutoff 150 m [7], and conditional Hungarian-matched position RMSE. E-OSPA is averaged over sensor-time pairs; RMSE averages their finite matched-set RMSE values, excluding a nonempty set paired with an empty set. The latter must be interpreted alongside absolute cardinality error. Focus consistency is mean pairwise E-OSPA between one fixed representative per formation over $t = 40$ –140, not an all-node agreement measure. Communication cost is attempted posterior-payload bytes, including failed transmissions, but excluding routing-control traffic.

4.2. What the comparisons establish

Table 1 separates repair from sparsification. Full causal repair improves the three reported error measures over fixed routing at both scales, but increases attempted bytes by 10.052% on M24 and 5.708% on X36. The sparse policy instead saves 10.041% and 21.830% of attempted bytes. On M24 its E-OSPA and conditional RMSE improve by 2.403% and 46.190%, and focus consistency improves by 1.449%. However, the worst formation-wise RMSE change is a 24.085% degradation.

On X36 sparse repair improves E-OSPA by only 0.368% despite a 47.655% conditional-RMSE reduction. Mean absolute cardinality error rises from 18.456 to 18.597, and focus consistency worsens by 0.776%. Consequently, the RMSE decrease cannot be interpreted as a comparable improvement in complete target-set recovery. The large cardinality errors also show that the global target-set task remains difficult under limited views and finite communication.

The RMSE averages also use different finite-cell sets: X36 sparse repair has finite RMSE on 97.465% of sensor-time cells, versus 100% for fixed routing. On their common finite cells, the means are 19.329 and 31.394 m, a 38.431% reduction. This comparison aligns sensor-time cells, not necessarily matched target identities.

Table 1. Complete paired development episodes, seed 1301. Lower values are better in all four metric columns. Error measures are in metres; bytes are decimal MB of attempted posterior payload. Messages are scheduled, before delivery. Each scale has its own fixed baseline; no confidence intervals can be inferred from a single seed.

Scale	Routing policy	E-OSPA	Cond. RMSE	Focus consistency	Payload	Messages/step
M24	Fixed formation tree	125.478	22.640	133.599	40.769	46–48
	Full causal repair	122.380	14.081	131.913	44.867	48
	Sparse causal repair	122.462	12.183	131.664	36.676	30
X36	Fixed formation tree	132.680	36.925	139.407	76.871	66–72
	Full causal repair	131.795	19.586	139.004	81.259	72
	Sparse causal repair	132.192	19.329	140.489	60.090	46
	Sparse + self fallback	131.961	20.263	139.273	60.317	46

Table 2. Sparse-arm count-error budget on the same saved episodes. The share is $\sum C / \sum E^2$, not a linear decomposition of mean OSPA. The final column bounds mean OSPA reduction with each cell’s count fixed; it is not an achieved tracking gain.

Scale	Count share	Ambiguous	Headroom
M24	99.567%	0/3840	0.279 m
X36	98.074–99.337%	121/5760	1.311 m

4.3. How much can localization alone improve?

To separate position refinement from target-set recovery, write squared OSPA at one sensor–time cell as

$$E^2 = C + L, \quad C = c^2 \frac{|n - T|}{\max(n, T)}, \quad L \geq 0, \quad (9)$$

where T, n are true and estimated cardinalities and $c = 150$ m. Set $E = C = L = 0$ when both sets are empty. If the estimated count is held fixed, zero localization error still leaves the floor $\sqrt{C}$. The saved summaries record $d = |n - T|$, rather than signed counts. For each nonnegative $n \in \{T - d, T + d\}$, the recorded matched RMSE R supplies a feasible-assignment upper bound

$$C \leq E^2 \leq C + \frac{\min(n, T)}{\max(n, T)} \min(R^2, c^2). \quad (10)$$

The OSPA and RMSE assignments need not coincide: the latter still supplies a feasible OSPA assignment. Nonempty–empty comparisons have $E = c$ and undefined R . We remove impossible counts and retain both possibilities when ambiguous, yielding $C_{\min} \leq C \leq C_{\max}$.

Table 2 shows that most error is in the count term. The localization-only headroom is at most $\text{mean}(E - \sqrt{C_{\min}})$, whereas useful recovery of missing targets can change C itself. This is not causal evidence about a particular edge or proof that the emitted estimates represent correct targets. It identifies the next design ques-

tion: whether existence evidence fails to arrive or fails to survive the subsequent fusion.

A packet-only replay using the same delivery uniforms finds that the sparse graph is strongly connected after delivery on 41/160 M24 steps and 18/160 X36 steps. Planned matrices are non-doubly-stochastic on 17 and 22 steps, whereas packet-renormalized matrices are non-double on 122 and 143 steps. This replay assumes that a delivered packet supplies an input; empty payloads and label-specific censoring can further restrict fusion. The counts do not prove absence of connectivity over longer windows or identify the cause of the tracking differences.

4.4. Receiver ablation and limitations

Self fallback reduces X36 E-OSPA by 0.175% and focus disagreement by 0.865% relative to sparse renormalization, with cardinality error falling from 18.597 to 18.518. However, conditional RMSE worsens by 4.836%, including a 37.361% loss in the most affected formation. It fails the predefined 1% network-RMSE tolerance for an M24 follow-up. Relative to fixed routing, its four network means improve, but E-OSPA and consistency gains remain small (0.542% and 0.096%). Thus missing-input handling changes the tradeoff without establishing a uniformly preferable rule.

Next, transport and post-fusion existence loss must be distinguished. No statistical significance, control-inclusive byte saving, decentralized routing implementation or cross-layout validity is established. Independent seeds and a non-radial zipper-merge scene remain necessary.

5. CONCLUSION

Two development cases show lower posterior traffic, but target-set and formation-tail tradeoffs remain. Self fallback does not resolve them; reliable target-set recovery is the next objective.

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6. REFERENCES

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